

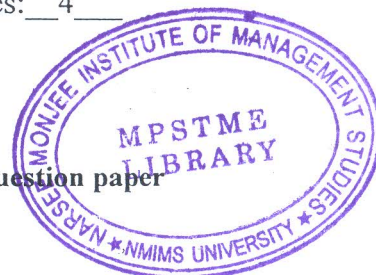
Program: B.Tech/MBA Tech Stream ;Data Science

Year: II Semester: IV

Subject: Statistical Structures in Data and Inference

Time: 3 hrs (10 am to 1 pm)Date: 18 / 04 / 24No. of Pages: 4Marks: 100**Final/Re Examination(2022-23)**

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.



- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO- ; 2 PO- ;1 BL-1	a.	Explain the use of Two-way ANOVA with a suitable example.	[5]
CO- ;1 PO- ;1 BL-1	b.	What is Type I and Type II Error.	[5]
CO- ; 2 PO- ;1 BL-1	c.	Explain how mathematically the standard deviation of the normal distribution is calculated through maximum likelihood estimation.	[5]
CO- ;2 PO- ;2 BL-1	d.	Explain R^2 in Linear Regression.	[5]
Q2			[20]
CO-3; PO-;1 BL-4	A	#Python Libraries <pre>import numpy as np from scipy.optimize import minimize from scipy.stats import norm</pre> Following are the sample observations for a random variable (normally distributed) 25,22,27,26,21,32,29,19,31,34 Write a Python program to find the most likely values for the mean and standard deviation through maximum likelihood estimation. Write down the python code in the answer book.	[10]

CO-2; PO-2 BL-33	B	#Python Libraries <pre>import numpy as np from scipy import stats</pre> Given a sample mean of 83, a sample standard deviation of 12.5, and a sample size of 22, write a Python program to test the hypothesis that the value of the population mean is 70 against the alternative that it is more than 70. Use the 0.025 significance level. Write down the python code in the answer book.	[10]
Q3			[20]
CO-3; PO-;10 BL-3	A.I	Let's assume that 30 out of 70 people recommend KFC to their friends. To test this claim, a random sample of 150 people obtained. Of these 150 people, 80 indicate that they recommend KFC to their friend. Is this claim accurate? Use $\alpha = 0.05$ to test the hypothesis. Use python code to do testing of hypothesis. Write down the python code in the answer book.	[10]
CO-1; PO-;11 BL-3	B	#Python Libraries <pre>import pandas as pd df=pd.read_csv('iris.csv')</pre> In the above dataset(iris.csv) investigate whether iris species have different average Sepal lengths or not using ANOVA. Write down the python code in the answer book	[10]
Q4			[20]
CO-;3 PO-;2 BL-4	A	A training program was conducted to improve participant's knowledge of the python language. Data of Test Results were collected from a selected sample both before and after the python training program. <pre>before=[39,43,41,32,37,40,42,40,37,38] after=[42,45,42,43,40,44,40,43,41,40]</pre> Test the hypothesis that the training is effective to improve participants' knowledge of python language at a 5% level of significance. Do a testing of hypothesis in python and cross check	[10]

		with pre-define library. Write down the python code in the answer book. (hint: pair t test)	
	B	The data set (profit.csv) consist of profit and different expenditure and Sate of 10 random companies in USA. Do a multiple regression in python and interpret the result. Write down the python code in the answer book. #Python Libraries import statsmodels.formula.api as smf	[10]
Q5			[20]
CO-;3 PO-;10 BL-4	A	A. Given the time between the arrival of two customers at a cafeteria is exponentially distributed with rate λ ($\lambda e^{-\lambda x}$), we have observed the time between the arrivals as $X=(4.23, 5.27, 7.84, 4.15, 3.45)$. For which value of λ is the probability of the observed sample the largest? (hint: likelihood estimation).	[10]
	B	Briefly explain multiple regression. Write a program (Python) to calculate regression coefficients to predict Rating based on Income, Limit, Cards and Balance (Data set: Credit.csv). Write down the python code in the answer book.	[10]
Q6			[20]
CO-;3 PO-;12 BL-4	A	How ridge regression is used to shrink estimates of the coefficients.	[10]
CO-;1 PO-;2 BL-1	B	import pandas as pd df=pd.DataFrame({'Years_of_Experience':[1,2,3,4,5,6,7,8,9,10], 'Salary':[48000,52000,65000,82000,115000,160000,210000,315000,512000,1100000]}) Build a best polynomial regression model on the above data set. Produce python code and analyze the output. Write down the python code in the answer book.	[10]
Q7			[20]

CO-;3 PO-;10 BL-4	A	A large hospital hires most of its nurses from the two major universities in the area. Over the last year, they have been giving a test to the newly graduated nurses entering the hospital to determine which school, if either, seems to educate its nurses better. Based on the following scores, help the personnel office of the hospital determine whether the schools differ in quality. Use the Mann–Whitney U test with a 10 percent level of significance. (P) <table><tr><th colspan="14">Test Scores</th></tr><tr><td>School A</td><td>97</td><td>69</td><td>73</td><td>84</td><td>76</td><td>92</td><td>90</td><td>88</td><td>84</td><td>87</td><td>93</td><td></td><td></td></tr><tr><td>School B</td><td>88</td><td>99</td><td>65</td><td>69</td><td>97</td><td>84</td><td>85</td><td>89</td><td>91</td><td>90</td><td>87</td><td>91</td><td>72</td></tr></table>	Test Scores														School A	97	69	73	84	76	92	90	88	84	87	93			School B	88	99	65	69	97	84	85	89	91	90	87	91	72	[10]
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CO-;1 PO-;2 BL-1	B	<pre>import statsmodels.api as sm from sklearn import linear_model import pandas as pd df3=pd.DataFrame({'x1':[2.78,2.44,2.09,2.14,2.72,2.65,4.92,4.37, 4.96,4.52,4.69,3.88], 'x2':[78,74,79,4,72,65,9,7,6,52,9,8], 'y':[0,0,0,0,0,0,1,1,1,1,1,1]})</pre> Build a logistic regression model on the above data set. Produce python code and analyze the output. Write down the python code in the answer book .	[10]																																										